

EXERCISES AND EXACT SOLUTIONS

Earn the Model's Complexity

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C04
Notebook	V7_0_N04
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add rolling-origin evaluation.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Compare MAE and RMSE consequences.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Recalibrate the interval.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Define a retirement trigger.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Refit/forecast at successive historical origins and aggregate horizon-specific metrics.

Solution 2

RMSE penalizes large errors more; choose according to decision loss rather than habit.

Solution 3

Estimate residual quantiles on a calibration window and verify held-out coverage and width.

Solution 4

Examples: persistent baseline underperformance, failed coverage/calibration, unacceptable subgroup harm, unmaintainable dependencies, or changed purpose.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated